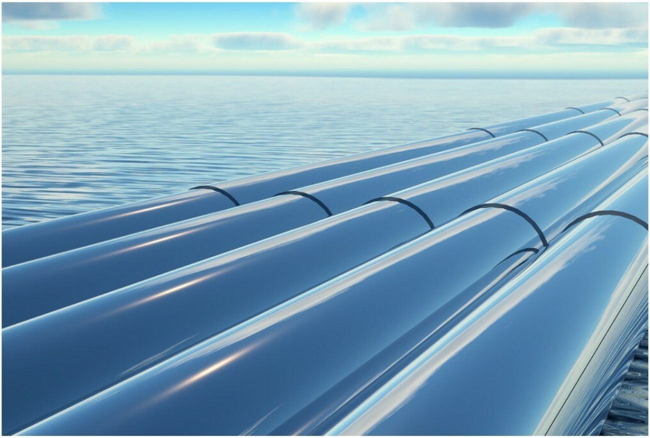
Using Jenkins to Create a Pipeline for Android Applications

This article is a tutorial on how to create a pipeline to perform code analysis using lint and create a APK file for Android applications.



Continuous integration is a practice that requires developers to integrate code into a shared repository such as GitHub, GitLab, SVN, etc, at regular intervals. This concept was meant to avoid the hassle of later finding problems in the build life cycle. Continuous integration requires developers to have frequent builds. The common practice is that whenever a code commit occurs, a build should be triggered. However, sometimes the build process is also scheduled in a way that too many builds are avoided. Jenkins is one of the most popular continuous integration tools.

Jenkins was known as a continuous integration server earlier. However, the Jenkins 2.0 announcement made it clear that, going forward, the focus would not only be on continuous integration but on continuous delivery too. Hence, 'automation server' is the term used more often after Jenkins 2.0 was released. It was initially developed by Kohsuke Kawaguchi in 2004, and is an automation server that helps to speed up different DevOps implementation practices such as continuous integration, continuous testing, continuous delivery, continuous deployment, continuous notifications, orchestration using a build pipeline or Pipeline as a Code.

Jenkins helps to manage different application lifecycle management activities. Users can map continuous integration with build, unit test execution and static code analysis; continuous testing with functional testing, load testing and security testing; continuous delivery and deployment with automated deployment into different environments, and so on. Jenkins provides easier ways to configure DevOps practices.

The Jenkins package has two release lines:

- *LTS (long term support):* Releases are selected every 12 weeks from the stream of regular releases, ensuring a stable release.